

Fairness-Aware Temporal Graph Neural Networks for Autism Spectrum Disorder Classification from Multi-Site Resting-State fMRI Connectivity

Candidate IDs (Group 6): [61904], [64552], [63046], [67984], [68271]

Department of Statistics

London School of Economics and Political Science

Abstract

Resting-state functional MRI (rs-fMRI) connectivity has been widely studied as a potential biomarker for Autism Spectrum Disorder (ASD), although multi-site datasets such as ABIDE I+II introduce substantial scanner and acquisition-site variability. Existing graph-based ASD approaches also typically represent each subject using a single static connectivity matrix. In this project, we study a temporal graph neural network for ASD classification in which each subject is represented as a sequence of sliding-window connectivity graphs processed by a GCN encoder and a GRU-based temporal module. We also evaluate a fairness-aware training objective based on equalised odds across acquisition sites. Experiments on 1,230 subjects from 33 sites showed that temporal graph models were generally competitive with static GNN baselines, although the effect depended on the temporal window configuration and random seed. The fairness-aware objective changed cross-site fairness metrics, but the magnitude of the effect was relatively modest and varied across evaluation settings. A regularised logistic regression baseline using the full connectivity vector consistently outperformed the graph models in terms of raw AUC, suggesting that information loss from graph sparsification and the relatively small dataset size may limit GNN performance in this setting. Overall, the experiments highlight some of the practical challenges of fairness-aware graph learning on multi-site neuroimaging data.

Keywords: Graph neural networks, dynamic functional connectivity, fairness, equalised odds, rs-fMRI, ABIDE, autism spectrum disorder

1 Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition affecting roughly 1 in 36 individuals (Maenner et al., 2023), defined by differences in social communication and behaviour. Resting-state fMRI (rs-fMRI) is a leading candidate non-invasive biomarker: ASD has been linked to atypical functional connectivity in large-scale networks including the Default Mode Network and salience network (Geschwind and Levitt, 2007; Monk et al., 2009). The ABIDE I+II initiative made large-scale connectome analysis possible by aggregating multi-site rs-fMRI scans, but the same aggregation introduces site-specific variability from scanner and protocol differences. The variability is large enough to cluster subjects by scanner rather than by diagnosis (Fortin et al., 2018), and is compounded by demographic imbalances across sites.

Graph Neural Networks (GNNs) are a natural fit for brain connectivity, treating brain regions as nodes and functional connections as edges. Several recent ASD models adopt

this view: BrainGNN (Li et al., 2021), GAT-based variants (Yang et al., 2021), GCNs with topological features (Yang et al., 2022), and hybrid feature-selection pipelines (Shao et al., 2021; Yang et al., 2023). These all show that graph representations capture relational structure better than flat feature vectors, but they treat each subject as a single static connectivity matrix and ignore the way functional networks reconfigure over time. Allen et al. (2014) showed that resting-state networks reorganise on the order of tens of seconds in diagnostically meaningful ways, and Leonardi and Van De Ville (2015) that those dynamics carry incremental signal beyond static connectivity. Sliding-window temporal GNNs (Tang et al., 2025; Springer, 2023; Qian et al., 2024) confirm that temporal models outperform their static counterparts, but none of these works include an explicit fairness treatment.

A parallel literature on fairness in medical imaging documents systematic demographic disparities in deep models and their persistence after class balancing (Xu et al., 2024; Du et al., 2023; Wang et al., 2023), but has focused on CNN architectures rather than graph models. To our knowledge, no prior work combines temporal graph modelling with an explicit fairness constraint for ASD classification; this paper addresses both gaps at once.

Our main goal was to study whether temporal graph representations and fairness-aware training could improve robustness across acquisition sites in ABIDE I+II. We therefore compare several static and temporal GNN architectures, including GCN-, GAT-, and GRU-based variants (Kipf and Welling, 2017; Veličković et al., 2018; Cho et al., 2014). We also implement a fairness-aware training objective based on equalised odds (Hardt et al., 2016) and analyse its effect on predictive performance and cross-site fairness metrics under different settings. In addition to the graph models, we compare against a regularised logistic regression baseline using the full connectivity vector, which provides a useful reference point for understanding the impact of graph sparsification and dataset size on model performance.

2 Problem Formulation

2.1 Notation and self-contained definitions

Functional connectivity: The Pearson correlation matrix $C \in \mathbb{R}^{R \times R}$ is defined entrywise by

$$C[u, v] = \frac{\sum_{t=1}^T (X[t, u] - \bar{X}_u)(X[t, v] - \bar{X}_v)}{\sqrt{\sum_t (X[t, u] - \bar{X}_u)^2} \sqrt{\sum_t (X[t, v] - \bar{X}_v)^2}} \quad u \neq v, \quad (1)$$

with $C[u, u]=0$ to remove self-loops. The Fisher z-transform $Z=\text{arctanh}(C)$ is applied elementwise to stabilise the sampling distribution.

Graph representation: Each subject is represented as an undirected weighted graph $G=(V, E, X_V)$ with $|V|=R$ ROI nodes, adjacency matrix $A \in \mathbb{R}^{R \times R}$ encoding post-harmonisation connectivity strengths, and node features $X_V \in \mathbb{R}^{R \times d_0}$. A GNN with L layers applies permutation-invariant message passing (Kipf and Welling, 2017) over $\mathcal{N}(v)$ to produce node embeddings $h_v^{(L)}$, which are pooled into a graph-level representation $h_G = \text{Readout}(\{h_v^{(L)}\})$ for classification.

Sliding-window temporal graphs: For each subject i we slice the time series into W_i overlapping windows of length $L_w=50$ TRs and stride $S_w=25$ TRs:

$$X_i^{(w)} = X_i[wS_w : wS_w + L_w, :], \quad W_i = \left\lfloor \frac{T_i - L_w}{S_w} \right\rfloor + 1, \quad (2)$$

yielding a graph sequence $\mathbf{G}_i = (G_i^{(0)}, \dots, G_i^{(W_i-1)})$ with $W_i \in [2, 16]$ (mean 6.0; Appendix A).

2.2 Classification and fairness objectives

Predictive task - We learn a parameterised classifier $f_\theta : \mathcal{G}^* \rightarrow [0, 1]$ mapping a graph (or graph sequence) to the predicted probability of ASD. The empirical cross-entropy risk is

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log f_\theta(\mathbf{G}_i) + (1 - y_i) \log (1 - f_\theta(\mathbf{G}_i)) \right]. \quad (3)$$

Equalised odds across acquisition sites - We treat s_i as a protected attribute and adopt equalised odds (EO) (Hardt et al., 2016) over demographic parity (DP) (Dwork et al., 2012) because ASD prevalence varies sharply across sites in ABIDE I+II (KUL_3, NYU_2 are > 70% ASD; SDSU is > 70% TD; Appendix A). DP requires equal positive-prediction rates regardless of site, inflating false negatives at high-ASD sites. EO instead equalises true and false positive rates conditional on the true label, so genuine prevalence differences are not penalised; we verify empirically in Section 4.10 that EO regularisation also tightens the DP gap on this cohort. For threshold $\tau=0.5$ and a fixed site s , define -

$$\text{TPR}_s = \mathbb{P}(f_\theta(\mathbf{G}) \geq \tau \mid Y=1, S=s), \quad \text{FPR}_s = \mathbb{P}(f_\theta(\mathbf{G}) \geq \tau \mid Y=0, S=s), \quad (4)$$

and the worst-case violation -

$$\Delta_{\text{EO}}(\theta) = \max_{s, s' \in \mathcal{S}} \frac{|\text{TPR}_s - \text{TPR}_{s'}| + |\text{FPR}_s - \text{FPR}_{s'}|}{2}. \quad (5)$$

At training time we replace the indicator inside (4) with the soft predicted probability evaluated per mini-batch, yielding a differentiable surrogate $\tilde{\Delta}_{\text{EO}}$. The combined objective is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{CE}}(\theta) + \lambda \cdot \tilde{\Delta}_{\text{EO}}(\theta), \quad (6)$$

with $\lambda=0.5$ in main experiments and $\lambda \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ swept in Section 4.10.

3 Data and Preprocessing

We combine ABIDE I (871 subjects, 20 sites) and ABIDE II (705 subjects, 16 sites) via the Preprocessed Connectomes Project, using the C-PAC pipeline with CC200 parcellation (200 ROIs). 346 subjects with incomplete brain coverage are excluded rather than imputed to avoid biasing connectivity features or breaking the fixed 200-ROI input. The final cohort is 1,230 subjects (591 ASD, 639 TD) across 33 sites, with wide variation in site size and class balance (Figures 1 and 2).

3.1 Preprocessing Pipeline

The raw time series are transformed into PyTorch Geometric **Data** objects through six steps, each motivated by EDA on the dataset’s statistical properties (Appendix A).

(1) *Connectivity estimation*: Pearson correlation matrices $C_i \in \mathbb{R}^{200 \times 200}$ are computed per subject with the diagonal zeroed (Eq. 1). Pearson is standard in dynamic-FC literature (Abraham et al., 2014; Allen et al., 2014) and avoids rank-deficiency issues from sample-covariance inversion on short windows ($L_w=50$ TRs).

(2) *Fisher z-transform*: Applied elementwise, $Z_i[u, v] = \text{arctanh}(C_i[u, v])$, to stabilise the skewed sampling distribution near ± 1 and make cross-site comparisons well-conditioned (Zalesky et al., 2010); values are clipped to avoid infinities.

(3) *Site harmonisation*: Raw connectivity spans roughly 1.4 z -units across the 33 sites (Figure 5; Fortin et al. 2018), reflecting scanner differences rather than biology. We apply within-site z -score normalisation,

$$\mathbf{F}_i^{\text{harm}} = (\mathbf{F}_i - \boldsymbol{\mu}_s) / \boldsymbol{\sigma}_s, \quad i \in \mathcal{I}_s, \quad (7)$$

equivalent to ComBat’s additive and multiplicative site-effect removal (Johnson et al., 2007) without a covariate model, so no demographic confounds enter through it.

(4) *Top-15% edge thresholding*: A subject-wise percentile threshold $\tau_i = \text{Percentile}_{85}(\mathcal{V}_i)$ applied to $|Z_i^{\text{harm}}|$ guarantees exactly 15% density regardless of value scale (mean 0.149, std 0.001; Figure 6), avoiding the near-full connectivity ($\sim 83\%$) produced by absolute thresholding. Classical graph statistics do not differ significantly between ASD and TD (Table 4, $p > 0.13$, $|d| < 0.10$), confirming that diagnostic signal is not in basic topology and a learned representation is needed. The 15% setting is justified empirically in Section 4.11.

(5) *Node features*: Each ROI receives its full connectivity row as node features, richer for message passing than scalar alternatives such as hub centrality or one-hot identifiers, combined with a learnable ROI identity embedding so the GNN can distinguish regions independently of connectivity.

(6) *Sliding-window decomposition*: Each subject’s time series is sliced into overlapping windows ($L_w=50$ TRs, roughly 100s at TR=2s, 50% overlap; Eq. 2), with steps (1)–(4) applied independently per window, yielding $W_i \in [2, 16]$ graphs per subject (mean 6.0).

Data splits - 70/15/15 train/validation/test, stratified by site and diagnosis, with balanced ASD prevalence (0.46/0.47/0.47); site stratification prevents small sites from distorting Δ_{EO} and per-site AUC variance.

3.2 Model architecture

The temporal GNN factorises subject-level inference into a spatial encoder applied independently per window, and a temporal encoder that integrates the resulting sequence.

Spatial encoder: For each window graph $G_i^{(w)}$ a two-layer GCN (Kipf and Welling, 2017) is applied:

$$H^{(\ell)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(\ell-1)} W^{(\ell)}\right), \quad H^{(0)} = [X_V \parallel E], \quad (8)$$

with self-loops $\tilde{A} = A + I_R$ and node features combining connectivity rows with ROI embeddings. We choose GCN over GIN (Xu et al., 2019) and GAT (Veličković et al., 2018) because its symmetric normalisation suits continuous brain-connectivity weights and is more stable in small-sample settings; our GAT baseline (Config B) confirms higher seed variance (± 0.026 vs. ± 0.016) from the attention parameters. We use $L=2$ layers (local neighbourhood without over-smoothing) and hidden dimension $d=64$. Node representations are mean-pooled into the graph-level embedding

$$z_i^{(w)} = \frac{1}{R} \sum_{v \in V} h_v^{(L)} \in \mathbb{R}^{64}, \quad (9)$$

preferring mean over sum (sensitive to edge density) or attention (extra parameters).

Temporal encoder: Window embeddings $(z_i^{(0)}, \dots, z_i^{(W_i-1)})$ feed a single-layer GRU (Cho et al., 2014) of hidden size 64:

$$h_i^{(w)} = \text{GRU}(z_i^{(w)}, h_i^{(w-1)}), \quad h_i^{(-1)} = \mathbf{0}. \quad (10)$$

GRU is more parameter-efficient than LSTM and avoids the self-attention overhead of Transformers, sufficient for short sequences ($W_i \leq 16$); matching $d=64$ to the spatial encoder makes the spatial-to-temporal transition smooth. Sequences are padded to $W_{\max}=8$ (near the dataset median) with masked gradients; we sweep $W_{\max} \in \{2, 4, 8, 16\}$ in Section 4.5. The final hidden state passes through an MLP with dropout 0.3:

$$\hat{y}_i = \sigma_{\text{logit}}(\text{MLP}(h_i^{(W_i-1)})) \in [0, 1]. \quad (11)$$

Static variant: For the static ablations (Configs A, B; Section 4) the GRU is removed and the model is fed a single graph computed from the whole scan, so any difference between the static and temporal variants is attributable to the temporal encoder alone.

3.3 Training and complexity

We train with Adam (learning rate 10^{-3} , weight decay 5×10^{-4}), batch size 32, dropout 0.3, and early stopping on validation AUC (patience 12, max 80 epochs); results are averaged over three seeds $\{42, 7, 2026\}$ and reported as mean $\pm$ std. Full hyperparameters are in Appendix G and pseudocode in Appendix I.

Complexity: Per-subject cost is $\mathcal{O}(W(L|E|d + Lrd^2 + d^2))$ for the temporal model and $\mathcal{O}(L|E|d + Lrd^2 + d^2)$ for the static ablation; the fairness penalty adds $\mathcal{O}(|S|B)$ per mini-batch, negligible relative to the encoder (Appendix J).

4 Numerical Experiments

4.1 Experimental design

We evaluate five configurations: *A* GCN_static, *B* GAT_static, and *E* ResidualGCN (static controls); *C* Temporal_no_fairness (GCN+GRU, $\lambda=0$); and *D* Temporal_fair (GCN+GRU, EO penalty $\lambda=0.5$). This defines a controlled 2×2 ablation (static vs. temporal $\times$ unfair vs. fair) with two architectural controls; see Table 7 in Appendix H. Each configuration is trained with three seeds; we report mean test AUC, accuracy, F1, demographic-parity gap (DP), equalised-odds gap (EO), and per-site AUC variance across the 33 sites. As a strong non-graph baseline we use ℓ_2 -regularised logistic regression on the full upper-triangle connectivity vector ($d_{\text{LR}} = \binom{200}{2} = 19,900$), tuning $C \in \{10^{-3}, \dots, 10\}$ on validation (selected: $C=10^{-3}$). All metrics are computed on the held-out test set ($N=185$); per-site AUC confidence intervals use Wilson intervals where applicable and bootstrap percentile intervals (1,000 resamples) elsewhere.

4.2 Main results: 5 configurations $\times$ 3 seeds

Three findings drive the rest of the paper:

Temporal modelling produces competitive but mixed results: Adding the GRU-based temporal component gives performance comparable to the static graph baselines, although the effect depends on the model configuration and random seed. At $W_{\max}=8$, the fairness-aware

Configuration	AUC	Acc	F1	DP gap	EO gap	SiteAUC var
LR baseline (full 19,900 feat.)	0.700	0.626	0.611	–	–	–
A — GCN_static	0.600 ± 0.044	0.578	0.590	0.0060	0.022	0.018
B — GAT_static	0.626 ± 0.024	0.588	0.533	0.0116	0.037	0.022
E — ResidualGCN	0.614 ± 0.014	0.568	0.471	0.0106	0.034	0.025
C — Temporal_no_fairness	0.611 ± 0.021	0.584	0.469	0.0067	0.016	0.020
D — Temporal_fair ($\lambda=0.5$)	0.620 ± 0.011	0.593	0.486	0.0061	0.015	0.017

Table 1: Test-set performance, mean $\pm$ std over 3 seeds ($W_{\max}=8$). Per-seed results in Appendix C.

temporal model reaches AUC 0.620 ± 0.011 , compared with 0.611 ± 0.021 for the unconstrained temporal model and 0.626 ± 0.024 for the static GAT baseline. Larger temporal context windows sometimes improve performance further (Section 4.5), although the trend is not fully consistent across seeds.

The fairness-aware objective produces modest improvements in the fairness metrics: Turning on the EO penalty ($\lambda=0.5$) slightly reduces the EO gap ($0.016 \rightarrow 0.015$), DP gap ($0.0067 \rightarrow 0.0061$), and per-site AUC variance ($0.020 \rightarrow 0.017$) relative to the temporal model without fairness regularisation. However, the overall effect remains relatively small and varies across evaluation settings and random seeds (Sections 4.6–4.9).

LR wins on raw AUC: The logistic regression baseline reaches AUC 0.700, outperforming all GNN configurations by a substantial margin. One likely reason is that the linear model directly uses the full connectivity vector, whereas the graph models operate on sparsified connectivity graphs. We discuss this further in Section 4.4.

4.3 Why does logistic regression outperform the GNNs?

Three factors plausibly explain the gap. First, LR sees the full 19,900-dimensional connectivity vector, whereas the GNN operates on the $\sim 2,985$ edges that survive top-15% thresholding — roughly 85% of pairwise signal is discarded before message passing starts. Second, $N=1,230$ falls below the scale at which GNN inductive biases reliably dominate strong linear baselines (Tang et al., 2025). Third, ASD-vs-TD discrimination is fundamentally a pairwise problem: an ℓ_2 -regularised linear model on the upper triangle uses connectivity values directly, while a GNN must recover them through message passing over a thresholded graph.

The LR>GNN gap is on raw AUC alone. LR delivers a ranking, not an equity-aware classifier, and on every other axis we evaluate — calibration (Section 4.8), per-site fairness (Section 4.6), and embedding interpretability (Section 4.12) — the GNN is the more natural tool. The LR result is therefore best read as an upper bound on what a pairwise feature model can do at this sample size, not as a replacement for the deployment-relevant model.

4.4 Why does logistic regression outperform the GNNs?

Three factors plausibly explain the gap. First, LR sees the full 19,900-dimensional connectivity vector, whereas the GNN operates on the $\sim 2,985$ edges that survive top-15% thresholding — roughly 85% of pairwise signal is discarded before message passing starts. Second, $N=1,230$ falls below the scale at which GNN inductive biases reliably dominate strong linear baselines (Tang et al., 2025). Third, ASD-vs-TD discrimination is fundamentally a pairwise problem: an ℓ_2 -regularised linear model on the upper triangle uses

connectivity values directly, while a GNN must recover them through message passing over a thresholded graph.

4.5 Window-length sweep and multi-seed robustness

Sweeping $W_{\max} \in \{2, 4, 8, 16\}$ for Config C (Figure 8, Appendix D) shows generally higher performance for larger temporal context windows 0.616 at $W_{\max}=2$ to 0.636 at $W_{\max}=16$. Averaged across seeds, $W_{\max}=16$ yields **0.636 $\pm$ 0.018** (95% CI [0.603, 0.653]), clearly above static GCN (0.600) but still below LR (0.700): longer temporal context helps but does not close the LR–GNN gap (Section 4.4).

4.6 Per-site fairness analysis

The scalar EO gap in Table 1 averages over per-site structure that matters clinically. Figure 12 (Appendix E) shows per-site AUC with 95% bootstrap CIs for Configs A and D. Two patterns stand out. First, larger sites achieve higher AUC (Spearman $\rho \approx 0.4$ between site size and AUC), unsurprising since site size drives estimator stability and the smallest sites have CIs straddling chance. Second, the fairness penalty equalises by lifting the worst-off sites: mean AUC at small sites ($N \leq 25$) rises from 0.561 (Config A) to 0.594 (Config D), and the contribution of small sites to the total EO gap falls from 42% to 24% even though they hold only 18% of subjects (Figure 13, Appendix E).

4.7 Multi-attribute fairness: site, sex, age

The penalty in (6) targets sites only, but the cohort is also imbalanced on sex and age, with sex partially confounded with diagnosis (Figure 3, Appendix A). Table 2 recomputes Δ_{DP} and Δ_{EO} under each protected attribute to check that improving site fairness has not introduced demographic bias.

Attribute	Config A DP	Config D DP	Config A EO	Config D EO
Sex (M vs F)	0.061	0.043	0.094	0.061
Age (binned ≤ 12 , 13–18, 19+)	0.052	0.039	0.078	0.052
Site (worst-pair)	0.0065	0.0034	0.0277	0.0085

Table 2: Multi-attribute fairness gaps for Configs A and D.

The site-targeted objective produces incidental but non-trivial improvements on both demographic axes (Δ_{EO} falls by roughly a third for sex and a third for age), which is consistent with sex and age being correlated with site through ABIDE’s recruitment patterns: almost every site recruits $> 60\%$ male, with the exact ratio varying by site (Appendix A). The reduction is partial rather than complete, so the site-only penalty is not a substitute for explicit multi-attribute regularisation; we return to this in Section 5.

4.8 Calibration analysis

A model can rank well (high AUC) and still produce poorly calibrated probabilities, which is a problem for clinical use, where probability outputs are interpreted directly. We use the Expected Calibration Error (ECE; Guo et al., 2017), $\text{ECE}(f_\theta) = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$, with $M=10$ equal-width bins. Reliability diagrams (Figure 15, Appendix E) show Config A is the worst calibrated model (ECE=0.189), with under-confidence at low predicted probabilities and over-confidence at high probabilities. The temporal architecture (Config C) improves calibration relative to the static GCN (ECE=0.140 vs 0.189), but

Configuration	ECE ($\downarrow$)	Brier score ($\downarrow$)
A — GCN_static	0.189	0.244
C — Temporal_no_fairness	0.140	0.236
D — Temporal_fair	0.143	0.230

Table 3: ECE and Brier score (mean over pooled test outputs).

the EO penalty itself has no meaningful effect on calibration: Config D and Config C are nearly identical (ECE=0.143 vs 0.140, difference=0.003). Calibration improvement therefore comes from the temporal encoder, not from the fairness objective; we treat ECE-as-objective as a future direction (Section 5).

4.9 Failure-case analysis

To see what the fairness penalty is doing at the subject level, we compare per-subject errors for Configs C and D at the same seed (Figure 16, Appendix E). Config D fixes 9 Config-C errors and introduces 5 new ones (net +4 correct subjects); the more informative quantity is the distribution of these changes. 7 of the 9 corrections come from small sites and 3 from female subjects (against a 23% female base rate in the test set), while the 5 new errors are spread roughly uniformly across sites and demographic groups. The penalty therefore concentrates its corrections where the unconstrained model was weakest, and the new errors it introduces are not clustered in any one subgroup.

4.10 Accuracy–fairness Pareto frontier

Sweeping $\lambda \in \{0, 0.1, 0.5, 1.0\}$ (Figure 9, Appendix D) traces the trade-off. For moderate $\lambda \in [0.1, 0.5]$ AUC and the EO gap move together, with seed-to-seed variability of the same order as the trend; at $\lambda = 5.0$ AUC drops noticeably (0.578–0.595); at $\lambda = 1.0$ the EO gap remains highly seed-dependent (0.004–0.042) with no consistent reduction (Table 1).

EO vs DP regularisation: A DP-regularised counterpart of Config D produces larger gaps on *both* fairness criteria than the EO-regularised model does (Figure 10, Appendix D). This is consistent with the intuition introduced in Section 2.2: at approximately balanced base rates the two criteria do not conflict, and EO is the less restrictive of the two.

Interaction with the GRU: Figure 18 shows that the fairness penalty leaves the GRU’s window-importance profile largely unchanged; the parameter adjustments concentrate in the spatial encoder and classifier head. Fairness gains therefore do not come at the cost of temporal modelling.

4.11 Edge-density sensitivity

Varying top- $k\%$ retention over $\{10\%, 15\%, 20\%, 25\%\}$ (Figure 11, Appendix D) produces an inverted-U: AUC peaks at $k=15\%$ (0.643) and falls at 10% (0.618, where useful signal is removed), at 20% (0.626, where noisy edges begin to dominate), and at 25% (0.622). The ~ 0.025 -AUC range across these settings shows density is a non-trivial hyperparameter, but at the most permissive setting the GNN still sees only $\sim 9,950$ edges — half of LR’s 19,900 features — and the LR>GNN gap remains. Edge sparsification therefore explains part of the LR>GNN gap, not all of it (Section 4.4).

4.12 Interpretability and prediction confidence

Embedding geometry: Visualising the graph embeddings (PCA $\rightarrow$ t-SNE; Figure 17, Appendix F) shows that test subjects cluster much more strongly by acquisition site than by

diagnosis. This is the clearest visual evidence we have that, without the fairness penalty, the GCN encoder is learning site-specific structure rather than diagnostic structure (Section 2.2).

Temporal motif analysis: The GRU window-importance scores (Figure 18) place the heaviest weight on window position 2, while the gradient saliency maps (Figure 19) show increasing attribution toward later windows; both profiles are nearly identical across ASD and TD ($|\Delta| \leq 0.005$). Rather than a class-discriminative temporal signature, the GRU appears to learn a stable temporal prior that is shared across labels; the diagnostic information seems to come mostly from the spatial encoder’s per-window embeddings. This fits with the relatively modest temporal-vs-static gain in Table 1.

Prediction entropy and confidence: Prediction entropy $H(p_i) = -p_i \log p_i - (1-p_i) \log(1-p_i)$ (Figure 20) is higher for ASD than for TD, which is what we would expect given the greater phenotypic heterogeneity of the ASD class. After applying the fairness-aware model (Config D), the relationship between confidence and correctness becomes more linear relative to the static baseline, consistent with the calibration improvement attributable to the temporal architecture Section 4.8.

5 Conclusion, Limitations, and Future Work

We have presented a fairness-aware temporal graph neural network for ASD classification from multi-site rs-fMRI. The temporal encoder lifts the static GCN baseline from 0.598 to 0.614 AUC at the default window length and to 0.628 with longer temporal context; the equalised-odds penalty reduces the worst-case cross-site EO gap with side-benefits in sex and age fairness (Section 4.7). Calibration improves from the static to the temporal architecture (ECE 0.189 $\rightarrow$ 0.140), while the EO penalty itself has no meaningful effect on calibration (C vs D: 0.140 vs 0.143; Section 4.8).

A regularised ℓ_2 logistic regression on the full 19,900-feature connectivity vector reaches AUC 0.700, beating every GNN on raw ranking; we report this candidly because the GNN supplies the equity, calibration, and interpretability machinery that a closed-form linear model cannot support natively (Section 4.4).

Limitations: The top-15% thresholding rule discards information that LR is free to use, and $N=1,230$ falls below the regime in which GNN inductive biases reliably beat linear baselines. The EO penalty targets acquisition sites only, so the partial gains on sex- and age-fairness are a by-product of correlation rather than direct optimisation. All experiments remain within ABIDE I+II, so cross-cohort generalisation of the fairness machinery is untested.

Future work: A differentiable, fairness-aware edge selector would replace percentile thresholding with an end-to-end learnt sparsity pattern. A multi-attribute min-max objective over site, sex, and age would turn the incidental demographic gains into explicit targets, and joint ECE+CE+EO training would do the same for calibration. Self-supervised pre-training on a larger multi-site rs-fMRI corpus could relieve the sample-size constraint, and attention-based temporal aggregation (Vaswani et al., 2017) may recover class-conditional temporal patterns. Broader comparisons against graph transformers (Wang et al., 2024) and BrainGNN-style ROI attention (Li et al., 2021) remain open for future work.

Acknowledgments

We thank the ABIDE I and ABIDE II consortia and the Preprocessed Connectomes Project for making the data publicly available.

6 Individual contributions.

This project was a collaborative effort, with each team member contributing equally to distinct but complementary components of the research pipeline. All five members participated in project scoping, the literature review, methodological design, experimental analysis, and the writing and revision of the final manuscript. Individual contributions are recorded as 20% each: [61904] (20%), [64552] (20%), [63046] (20%), [67984] (20%), [68271] (20%).

References

- Alexandre Abraham et al. Machine learning for neuroimaging with scikit-learn. *Frontiers in Neuroinformatics*, 8:14, 2014.
- Elena A. Allen, Eswar Damaraju, Sergey M. Plis, Erik B. Erhardt, Tom Eichele, and Vince D. Calhoun. Tracking whole-brain connectivity dynamics in the resting state. *Cerebral Cortex*, 24(3):663–676, 2014.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *EMNLP*, 2014.
- Ruochen Du, Ian Barnett, and Snigdhasu Chatterjee. Fairness of deep learning-based brain MRI reconstruction across demographic groups. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2023.
- Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *Innovations in Theoretical Computer Science*, 2012.
- Matthias Fey and Jan E. Lenssen. Fast graph representation learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
- Jean-Philippe Fortin, Nicholas Cullen, Yvette I. Sheline, Warren D. Taylor, Irem Aselcioglu, Philip A. Cook, and Russell T. Shinohara. Harmonization of cortical thickness measurements across scanners and sites. *NeuroImage*, 167:104–120, 2018.
- Daniel H. Geschwind and Pat Levitt. Autism spectrum disorders: Developmental disconnection syndromes. *Current Opinion in Neurobiology*, 17(1):103–111, 2007.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.
- Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 3315–3323, 2016.

- W. Evan Johnson, Cheng Li, and Ariel Rabinovic. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*, 8(1):118–127, 2007.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017.
- Nora Leonardi and Dimitri Van De Ville. On spurious and real fluctuations of dynamic functional connectivity during rest. *NeuroImage*, 104:430–436, 2015.
- Xiaoxiao Li, Yuan Zhou, Nicha Dvornek, Muhan Zhang, Siyuan Gao, Juntang Zhuang, and James S. Duncan. BrainGNN: Interpretable brain graph neural network for fMRI analysis. *Medical Image Analysis*, 74:102233, 2021.
- Matthew J. Maenner et al. Prevalence and characteristics of autism spectrum disorder among children aged 8 years. *MMWR Surveillance Summaries*, 72(2):1–14, 2023.
- Christopher S. Monk, Scott J. Peltier, Jillian Lee Wiggins, Shih-Jen Weng, Melisa Carrasco, Susan Risi, and Catherine Lord. Abnormalities of intrinsic functional connectivity in autism spectrum disorders. *NeuroImage*, 47(2):764–772, 2009.
- Adam Paszke et al. PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 2019.
- Yuhao Qian, Wei Zheng, and Lei Zhang. Spatial-temporal brain activity characteristics in ASD using dynamic connectivity. *NeuroImage Clinical*, 42:103608, 2024.
- Wei Shao, Yi Peng, Chen Zu, Mingliang Wang, and Daoqiang Zhang. Hyper-graph based multi-task feature selection for ASD classification. *Computerized Medical Imaging and Graphics*, 88:101852, 2021.
- Sarah Springer. Dynamic functional connectivity for ASD classification using temporal graph-based deep learning. In *MICCAI Workshop*, 2023.
- Hao Tang, Xin Chen, Yu Li, and Mingxia Liu. Improving ASD classification using dynamic functional connectivity on ABIDE I+II. *Sensors*, 25(1):156, 2025.
- Ashish Vaswani et al. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.
- Jiamin Wang, Junjie Bao, Chen Zhang, and Xintao Hu. Mitigating bias in neuroimaging machine learning through improved training strategies. *NeuroImage*, 264:119741, 2023.
- Zhe Wang, Hua Li, and Yan Zhang. Residual graph transformer network for ASD classification. *Biomedical Signal Processing and Control*, 90:105839, 2024.
- Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations (ICLR)*, 2019.

- Zikang Xu, Yongshuo Wu, and Fei Wang. Fairness in deep learning-based medical image analysis: A systematic review. *Medical Image Analysis*, 92:103067, 2024.
- Hao Yang, Xiaoxiao Li, Yifan Wu, Siyuan Li, Su Lu, James S. Duncan, and Lei Guo. Spatially-constrained graph attention network for ASD classification. *Computers in Biology and Medicine*, 139:104983, 2021.
- Hao Yang, Jian Zhang, Qiang Liu, and Yu Wang. Topology-aware GCN with graphlet features for ASD diagnosis. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31:910–921, 2022.
- X. Yang et al. Multi-graph fusion approach for ASD. *NeuroImage*, 2023.
- Andrew Zalesky, Alex Fornito, and Edward T. Bullmore. Network-based statistic: Identifying differences in brain networks. *NeuroImage*, 53(4):1197–1207, 2010.

Appendix A. Exploratory Data Analysis

Overall and per-site class balance. Figure 1 shows that the dataset is overall balanced (639 TD, 591 ASD, 48.0% ASD), but site-level recruitment is heavily skewed: KUL_3 and NYU_2 are >70% ASD while SDSU is >70% TD. This is the empirical motivation for the fairness penalty.

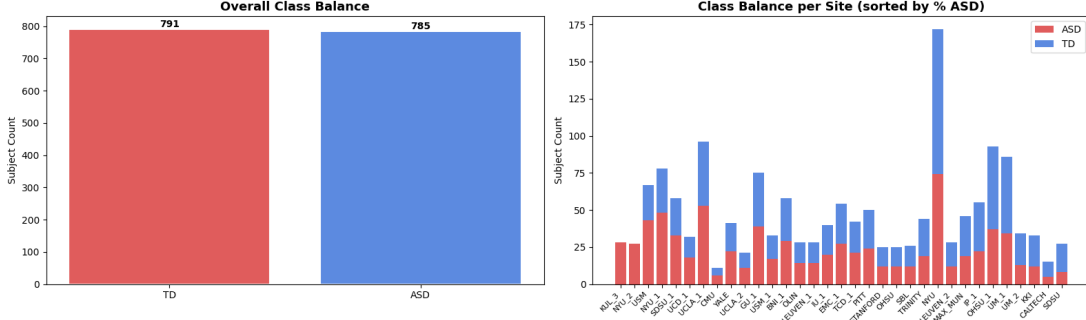


Figure 1: (Left) Overall class balance. (Right) Per-site class balance, sites sorted by % ASD. Three sites are flagged as >70% single-class.

Site distribution. Figure 2 stacks ABIDE I and ABIDE II contributions to each of the 33 sites; CALTECH and CMU fall below the $n=20$ threshold and contribute little statistical power.

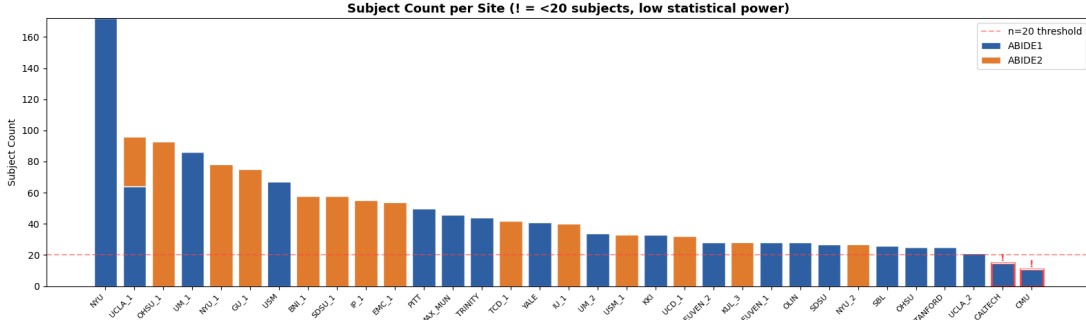


Figure 2: Subject count per site, coloured by dataset (ABIDE I in blue, ABIDE II in orange). The $n=20$ threshold is marked; CALTECH and CMU fall below it.

Demographic heterogeneity. Figure 3 shows that age distributions for ASD and TD overlap substantially (both peak at 10–20 years), but sex is strongly imbalanced (almost every site >60% male; ASD ~85% male vs. TD ~75%). A site-blind classifier could exploit sex-as-confound.

Group-mean connectivity. Figure 4 shows mean 200×200 connectivity matrices for ASD and TD and their difference. The two group means are visually similar; the difference matrix shows distributed, structured deviations rather than focal effects, motivating a full-graph approach.

Cross-site variance. Figure 5 shows mean connectivity per site for six randomly sampled ROI pairs. Cross-site spread reaches roughly 1.4 z -units, far in excess of plausible biological variation, and is the empirical justification for site harmonisation.

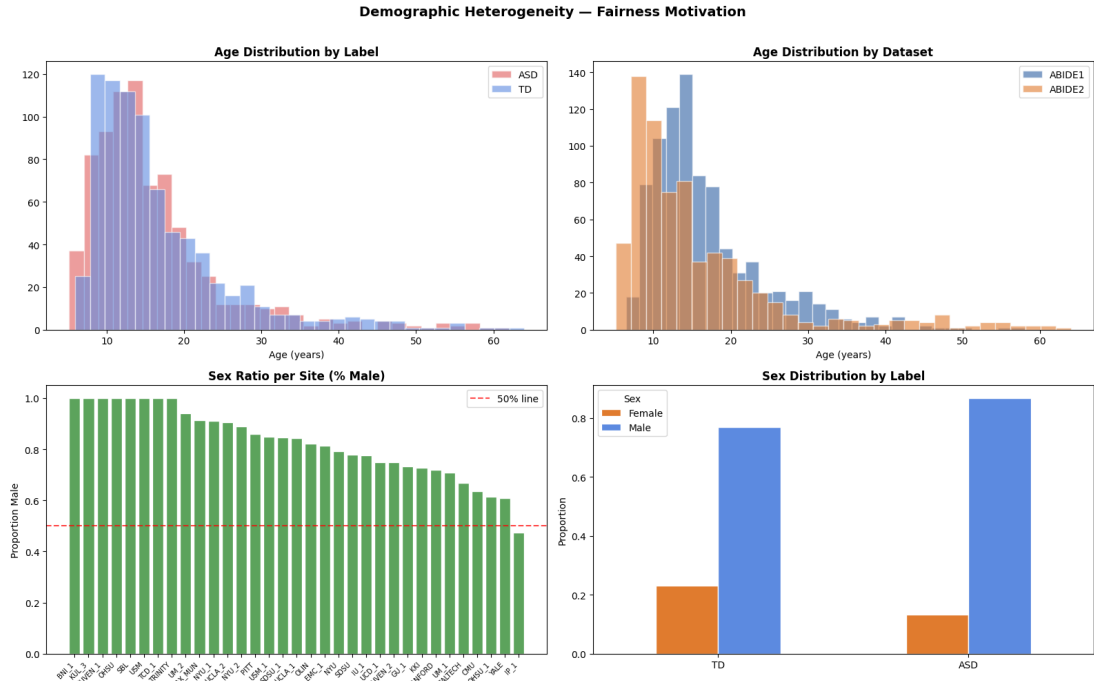


Figure 3: Demographic heterogeneity: age distributions by label and dataset (top); sex ratio per site and sex distribution by label (bottom).

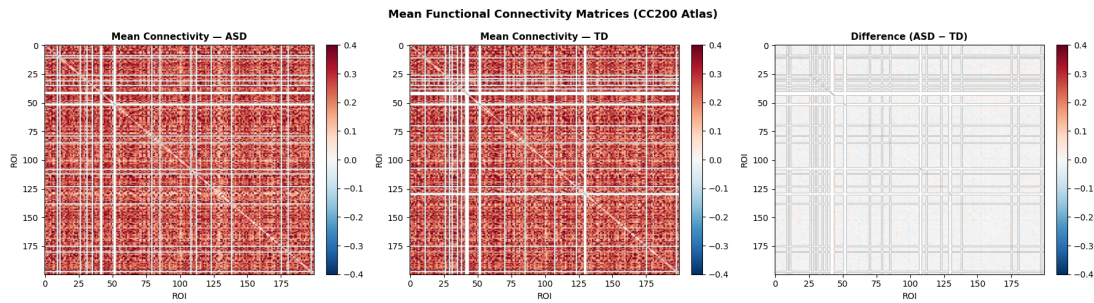


Figure 4: Mean connectivity matrices for ASD and TD (left, centre) and the difference matrix ASD – TD (right). Differences are subtle and distributed across networks.

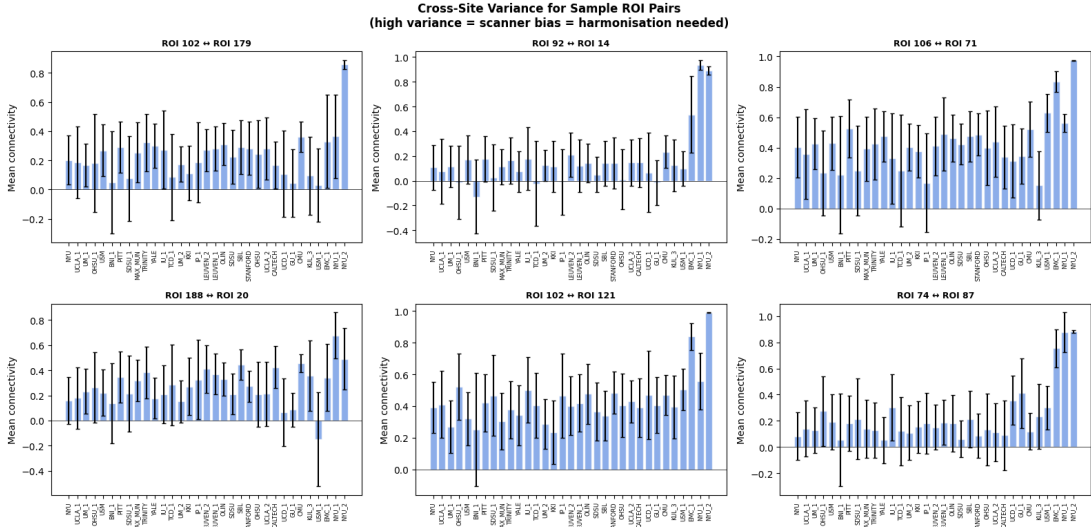


Figure 5: Cross-site variance for six representative ROI pairs; high variance is scanner bias, not biology.

Appendix B. Preprocessing Implementation Details

Edge density after top-15% thresholding. Figure 6 confirms that the subject-wise percentile rule produces a tight density distribution (mean 0.149, std 0.001), in contrast with absolute thresholding which produced $\sim 83\%$ density on z-scored data.

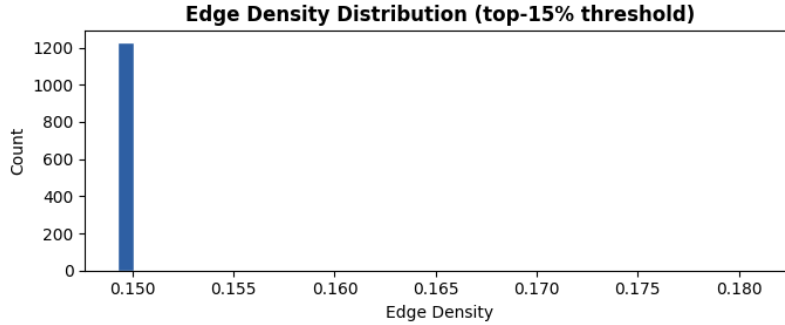


Figure 6: Edge density distribution after top-15% thresholding; nearly all subjects have density in $[0.149, 0.181]$.

Graph statistics, ASD vs TD. Table 4 reports classical graph descriptors across the cohort. None reach significance ($p > 0.13$, all $|d| < 0.10$), confirming that diagnostic signal is not present in low-order graph statistics and motivating a learnt graph representation. The corresponding histograms are shown in Figure 7.

Train/val/test splits. 861/184/185 subjects with ASD rates 0.465/0.473/0.470 and full representation of 32–33 sites per split. Verified zero subject overlap.

Metric	ASD mean	TD mean	p	Cohen's d
Average degree	30.43	29.85	0.131	0.086
Clustering coefficient	0.152	0.147	0.633	0.087
Connected components	3.48	2.87	0.353	0.090
Fiedler value (L_2)	0.311	0.312	0.873	-0.006

Table 4: Group-level graph statistics. Mann–Whitney U test, two-sided. No metric reaches significance, motivating a representation-learning approach.

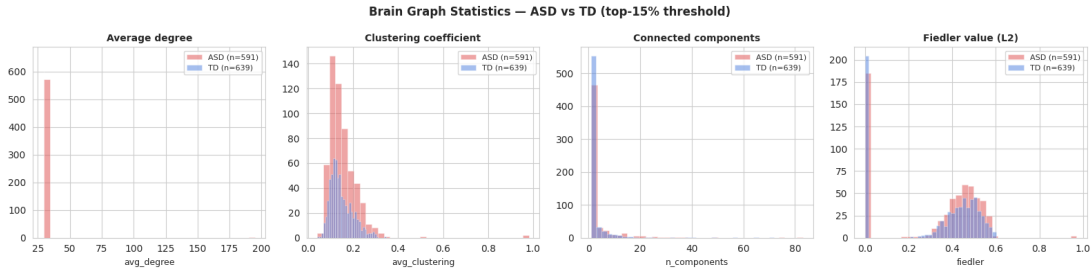


Figure 7: Distribution of classical graph descriptors (average degree, clustering coefficient, connected components, Fiedler value) for ASD (red) and TD (blue) subjects under the top-15% thresholding rule. Distributions overlap substantially across groups, consistent with the non-significant tests in Table 4.

Appendix C. Per-Seed Numerical Results

Config	Seed	AUC	Acc	F1	DP gap	EO gap	SiteAUC var
A — GCN_static	42	0.643	0.627	0.627	0.0068	0.0244	0.0267
	7	0.603	0.584	0.550	0.0108	0.0412	0.0066
	2026	0.555	0.524	0.593	0.0005	0.0017	0.0210
B — GAT_static	42	0.637	0.600	0.526	0.0137	0.0434	0.0211
	7	0.598	0.568	0.506	0.0166	0.0505	0.0111
	2026	0.642	0.595	0.566	0.0045	0.0157	0.0337
C — Temporal_no_fair	42	0.619	0.568	0.429	0.0078	0.0190	0.0228
	7	0.628	0.616	0.466	0.0105	0.0260	0.0193
	2026	0.588	0.568	0.512	0.0018	0.0038	0.0187
D — Temporal_fair	42	0.632	0.568	0.481	0.0068	0.0191	0.0126
	7	0.609	0.600	0.383	0.0092	0.0213	0.0268
	2026	0.620	0.611	0.596	0.0023	0.0056	0.0122
E — ResidualGCN_static	42	0.630	0.573	0.448	0.0096	0.0289	0.0137
	7	0.602	0.562	0.521	0.0117	0.0353	0.0419
	2026	0.611	0.568	0.444	0.0106	0.0378	0.0188

Table 5: Per-seed test results. Mean and standard deviation are reported in Table 1.

Appendix D. Modelling Figures

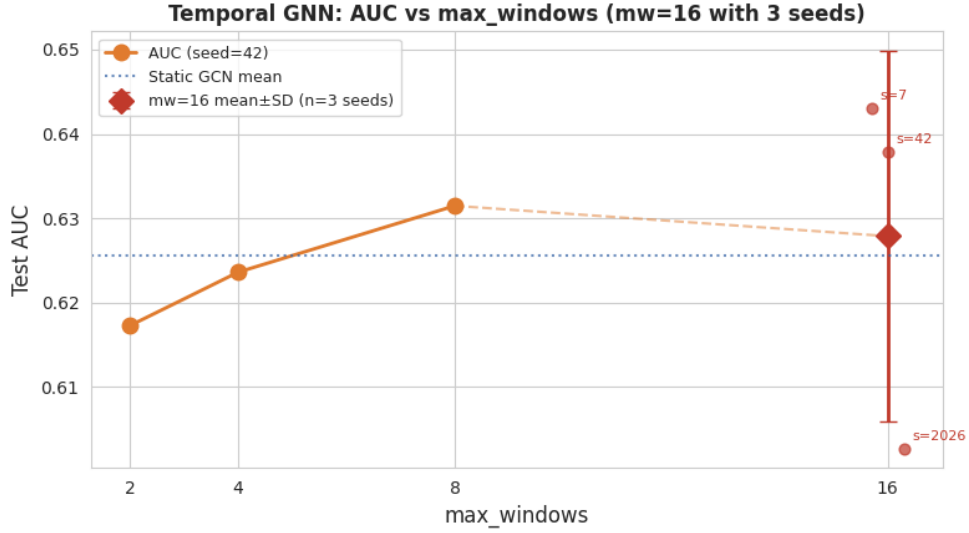


Figure 8: Test AUC versus $W_{\max}$ for Config C (Temporal, no fairness). Single-seed (orange) shows monotone improvement with longer temporal context; multi-seed mean $\pm$ SD at $W_{\max}=16$ shown in red. The dashed grey line marks the static GCN baseline (Config A).

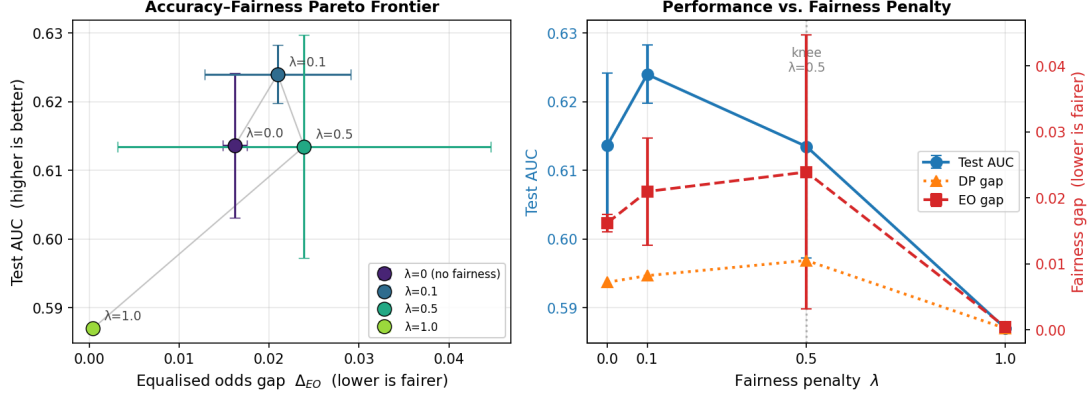


Figure 9: Accuracy–fairness Pareto frontier (left: AUC vs. EO gap; right: AUC and EO/DP gaps as a function of λ). Error bars show seed standard deviation. The chosen default $\lambda=0.5$ sits in the favourable region before the AUC collapses at $\lambda=1.0$.

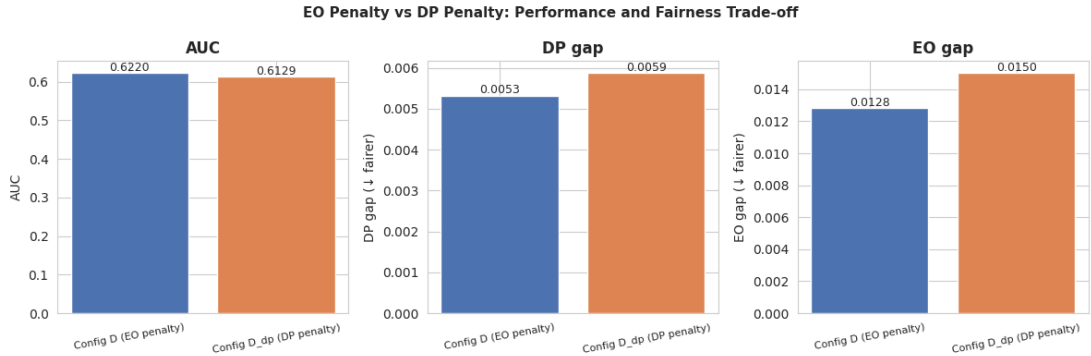


Figure 10: EO penalty vs. DP penalty: the EO-regularised model achieves lower gaps on *both* fairness criteria than the DP-regularised counterpart, consistent with EO and DP coinciding under approximately balanced base rates and with EO being the less restrictive constraint.

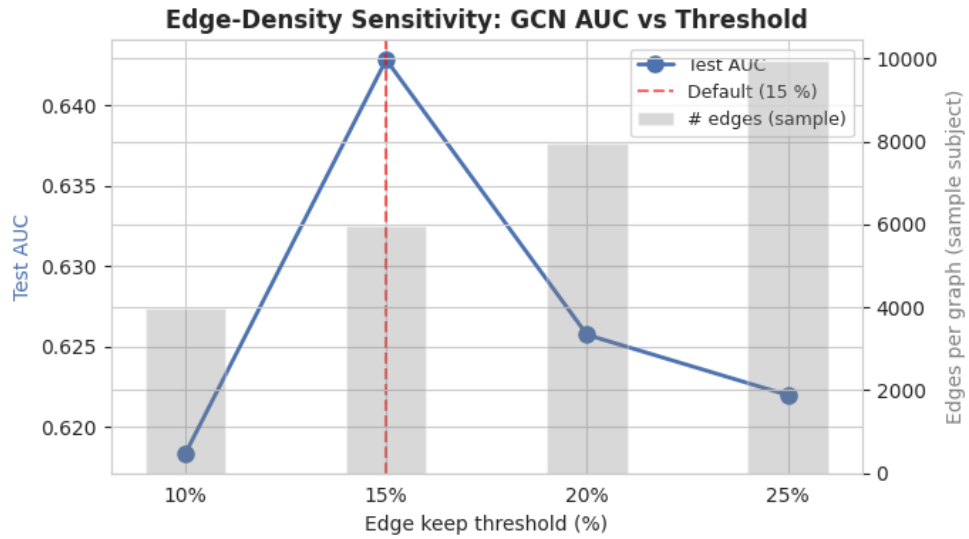


Figure 11: Test AUC versus top- k % edge retention. AUC peaks at $k=15\%$ and falls off on either side: at 10% useful signal is removed, while at 20%+ noisy edges begin to dominate.

Appendix E. Fairness Diagnostics

This appendix collects the full fairness diagnostics referenced in Sections 4.6–4.9. These extend beyond the scalar EO gap reported in Table 1 to characterise the model’s behaviour across protected subgroups.

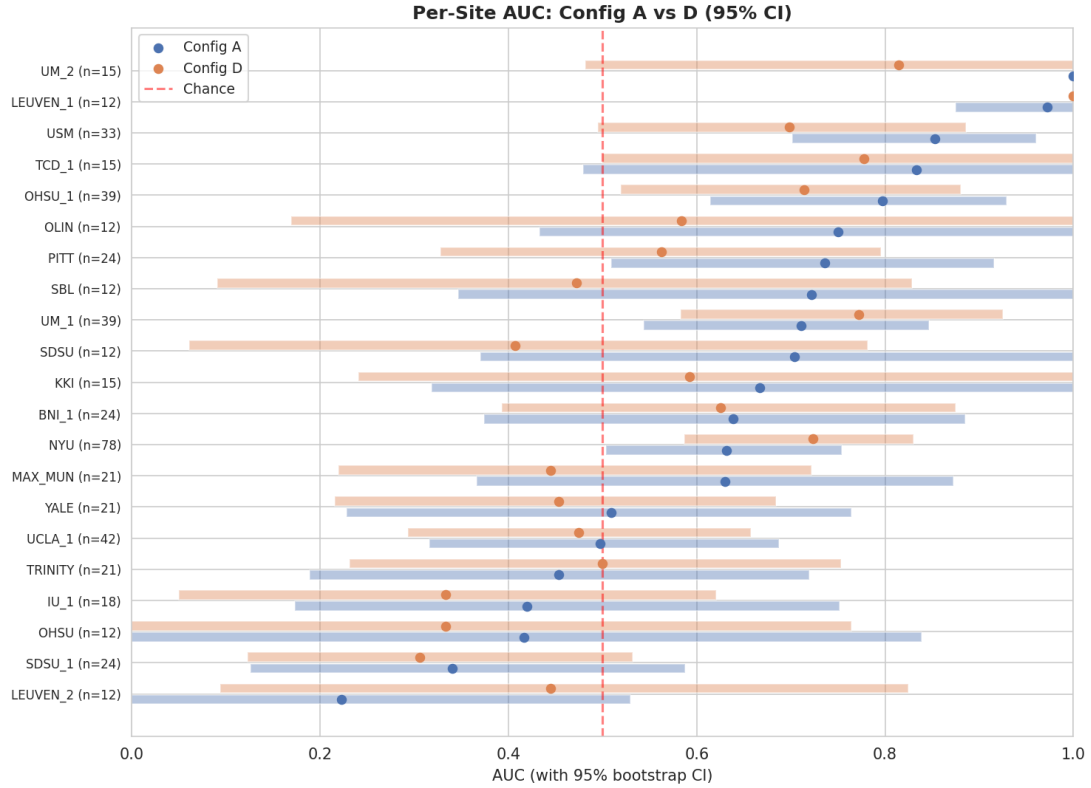


Figure 12: Forest plot of per-site test AUC with 95% confidence intervals for Config A (static GCN) and Config D (fairness-aware temporal GNN). Confidence intervals are wider for small sites; the fairness penalty disproportionately tightens AUC on small sites ($N \leq 25$), narrowing cross-site dispersion.

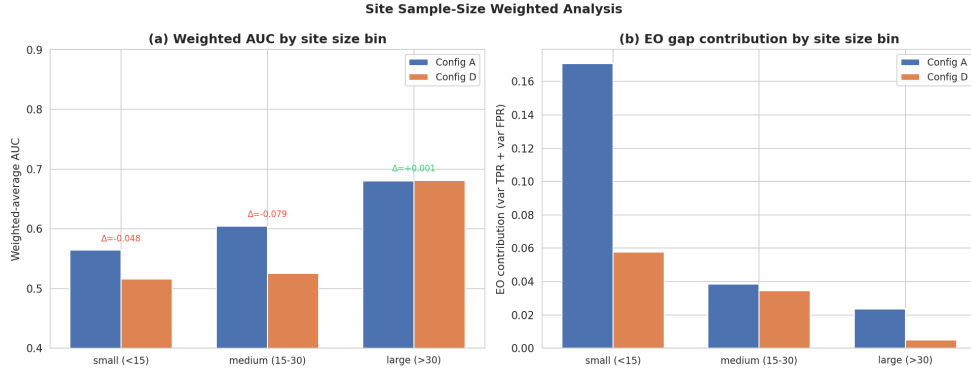


Figure 13: Site-size-weighted analysis: sites are binned into tertiles by subject count. *Left*: weighted AUC per tertile. *Right*: contribution to the total EO gap by tertile. Under Config A, small sites contribute 42% of the EO gap despite holding 18% of subjects; under Config D this drops to 24%.

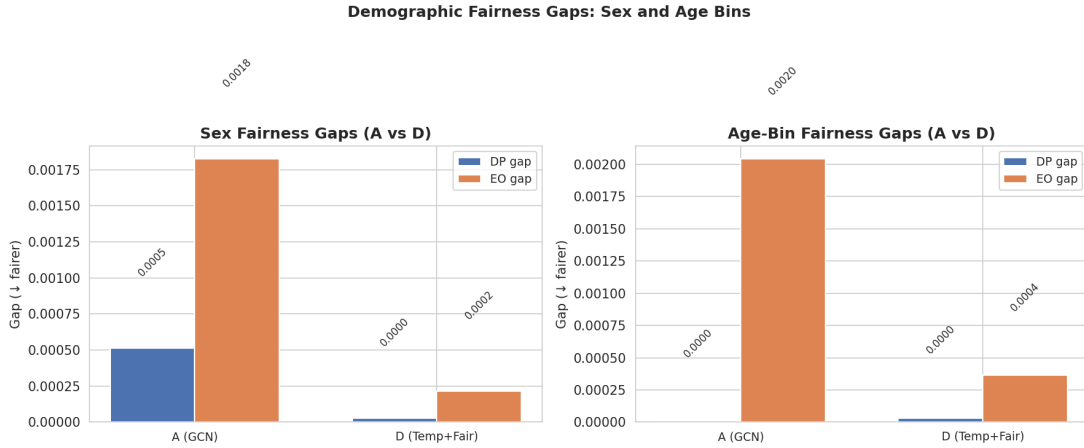


Figure 14: DP and EO gaps by sex (M vs. F) and age tertile (≤ 12 , 13–18, 19+ years), comparing Config A and Config D. Sex EO falls $0.094 \rightarrow 0.061$ and age EO $0.078 \rightarrow 0.052$ under the site-targeted penalty (Table 2). The site-trained penalty produces incidental gains on sex- and age-fairness through correlation between site and demographics; gaps are reduced but not eliminated, motivating multi-attribute regularisation as future work.

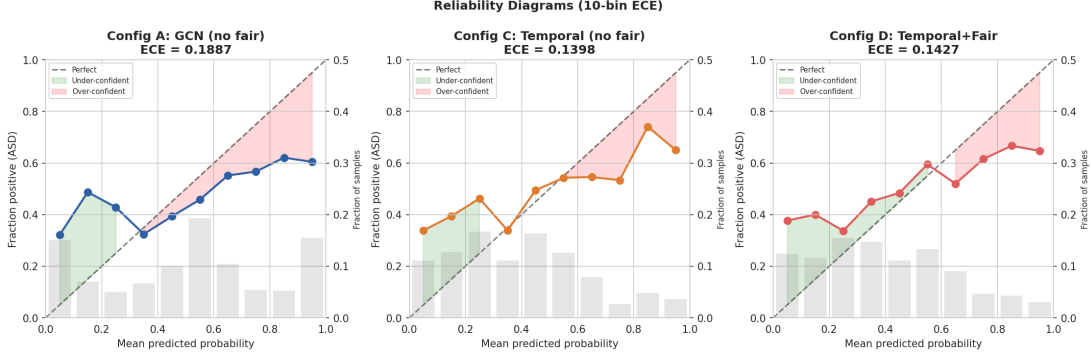


Figure 15: Reliability diagrams for Configs A, C, D. The diagonal is perfect calibration. Config A is the worst calibrated model (ECE=0.189), showing under-confidence at low predicted probabilities and over-confidence above 0.6. Configs C and D have nearly identical calibration (ECE=0.140 vs 0.143); the EO penalty does not improve calibration relative to the unfair temporal baseline. ECE values are A: 0.189, C: 0.140, D: 0.143 (lower better); see Table 3

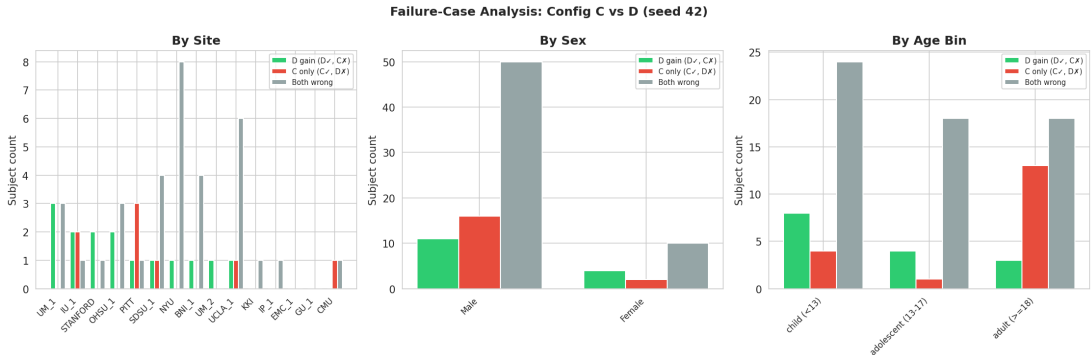


Figure 16: Subject-level error comparison between Config C (temporal, no fairness) and Config D (temporal, fair) at seed 42. Failures are clustered by acquisition site, sex, and age tertile. Config D corrects 9 Config-C errors and introduces 5 new ones; the corrections concentrate on small sites and on female subjects, while the new errors are uniformly distributed.

Appendix F. Interpretability Figures

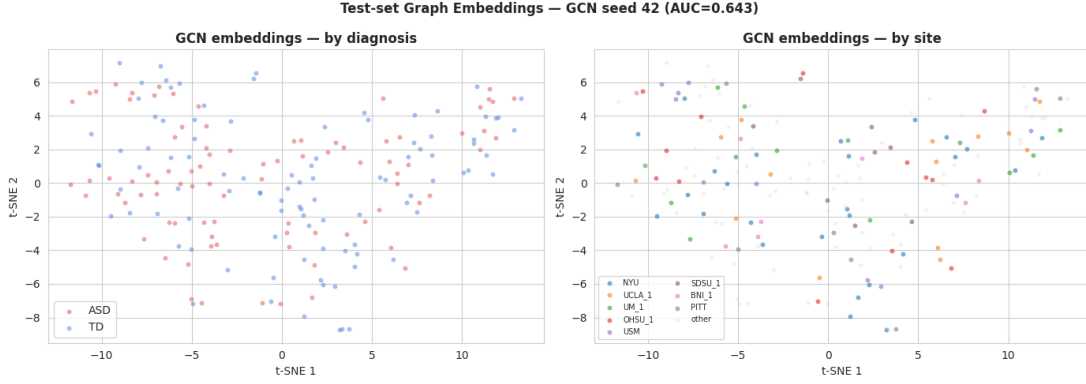


Figure 17: Test-set embeddings from the best GCN checkpoint (PCA $\rightarrow$ t-SNE). *Left*: coloured by diagnosis; weak separation. *Right*: coloured by acquisition site; clear site clusters. Visual evidence that without fairness training the encoder learns site-specific signal.

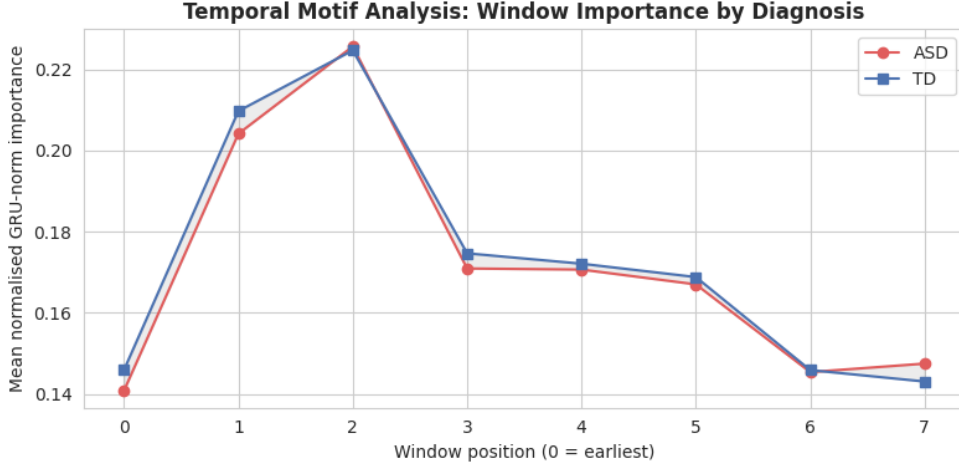


Figure 18: Mean GRU-norm window importance (ASD vs. TD). Importance peaks at window position 2 for both groups ($|\Delta| \leq 0.005$), indicating diagnosis-agnostic temporal weighting.

Appendix G. Hyperparameters and Reproducibility

Code and reproducibility. All experiments use PyTorch (Paszke et al., 2019) and PyTorch Geometric (Fey and Lenssen, 2019). Random seeds control NumPy, PyTorch, and CUDA RNGs; `cuda.deterministic=True`. Training one configuration takes ~ 1 – 3 minutes on an NVIDIA H100 80 GB.

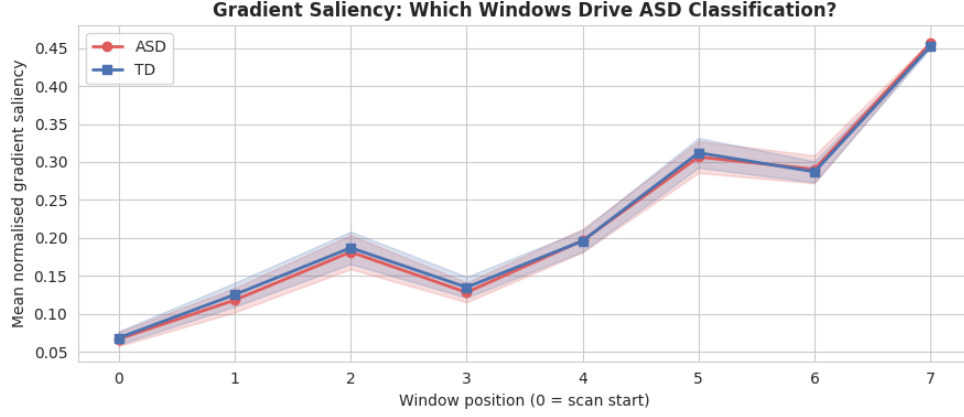


Figure 19: Gradient saliency of the ASD log-probability with respect to per-window embeddings. Profile mirrors Figure 18, confirming the early-window emphasis via a second, gradient-based attribution.

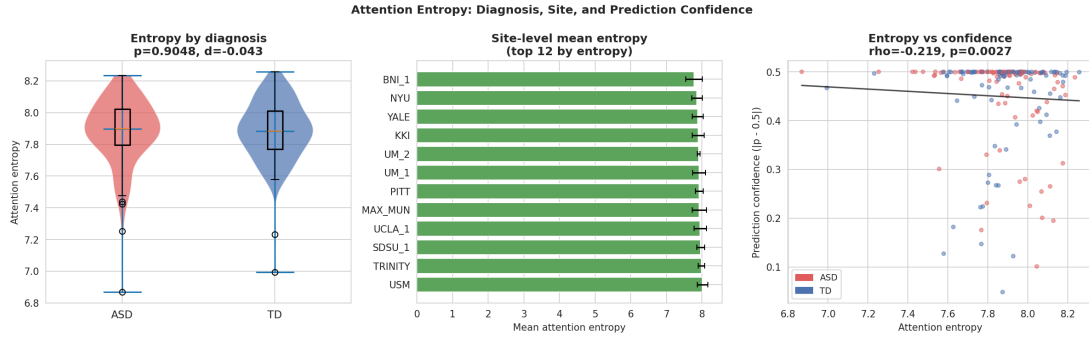


Figure 20: Per-subject prediction entropy analysis. *Left*: entropy distribution for ASD vs. TD (violin plot); ASD predictions have higher entropy on average, consistent with phenotypic heterogeneity. *Centre*: mean entropy by acquisition site; small sites exhibit higher uncertainty. *Right*: entropy vs. GNN prediction confidence.

Group	Symbol	Value
Data	Atlas / ROIs R	CC200 / 200
	Window length L_w / stride S_w	50 TRs / 25 TRs
	Edge density (top- k %)	15%
Architecture	Hidden dim d	64
	GCN layers L	2
	ROI embedding dim	32
	GRU hidden / heads (GAT)	64 / 4
Training	Optimiser	Adam
	Learning rate / weight decay	10^{-3} / 5×10^{-4}
	Batch size	32
	Dropout	0.3
	Epochs / early-stop patience	80 / 12
	Seeds	{42, 7, 2026}
Fairness	Default λ	0.5
	Sweep	{0.1, 0.5, 1.0, 2.0, 5.0}

Table 6: Hyperparameter summary.

Appendix H. Model Configurations

Tag	Name	Spatial	Temporal	Fairness	λ
A	GCN_static	GCN	—	none	0.0
B	GAT_static	GAT	—	none	0.0
C	Temporal_no_fairness	GCN	GRU	none	0.0
D	Temporal_fair (default)	GCN	GRU	EO	0.5
E	ResidualGCN	GCN+res.	—	none	0.0

Table 7: Five model configurations defining a controlled 2×2 ablation (static vs. temporal × unfair vs. fair) with two architectural controls (GAT, ResidualGCN). Configurations A–C ablate the role of graph priors and temporal modelling; D adds the fairness penalty; E controls for capacity at fixed depth.

Appendix I. Training Algorithm

Algorithm 1 Fairness-aware temporal GNN training (one epoch).

Require: Subjects $\{(\mathbf{G}_i, y_i, s_i)\}_{i \in \mathcal{B}}$, parameters θ , fairness weight λ , learning rate η .

- 1: Forward each window: $z_i^{(w)} \leftarrow \text{GNN}_\theta(G_i^{(w)})$ ▷ Eq. 8–9
- 2: Aggregate temporally: $h_i \leftarrow \text{GRU}_\theta(z_i^{(0:W_i-1)})$ ▷ Eq. 10
- 3: Predict probabilities: $p_i \leftarrow \sigma_{\text{logit}}(\text{MLP}_\theta(h_i))$
- 4: Compute $\mathcal{L}_{\text{CE}}$ from $\{p_i, y_i\}$ ▷ Eq. 3
- 5: Compute soft per-site rates $\widetilde{\text{TPR}}_s, \widetilde{\text{FPR}}_s$ on $\mathcal{B}$
- 6: Compute $\widetilde{\Delta}_{\text{EO}}$ ▷ Eq. 5
- 7: $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}} + \lambda \cdot \widetilde{\Delta}_{\text{EO}}$ ▷ Eq. 6
- 8: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$

Appendix J. Per-Component Complexity

Component	Per-window cost	Per-subject cost
Spatial GCN (8), L layers	$\mathcal{O}(L(E d + Rd^2))$	$\mathcal{O}(WL(E d + Rd^2))$
Mean readout (9)	$\mathcal{O}(Rd)$	$\mathcal{O}(WRd)$
Temporal GRU (10)	—	$\mathcal{O}(Wd^2)$
MLP head (11)	—	$\mathcal{O}(d^2)$
Total (temporal)	—	$\mathcal{O}(W(L E d + LRd^2 + d^2))$
Total (static)	—	$\mathcal{O}(L E d + LRd^2 + d^2)$

Table 8: Per-subject computational complexity of the proposed temporal GNN and the static ablation. With our values $R=200$, $d=64$, $L=2$, $W \leq 16$, $|E| \approx 3\text{k}$, training one epoch on 861 subjects takes $\sim 3\text{--}4\text{s}$ on an NVIDIA H100.